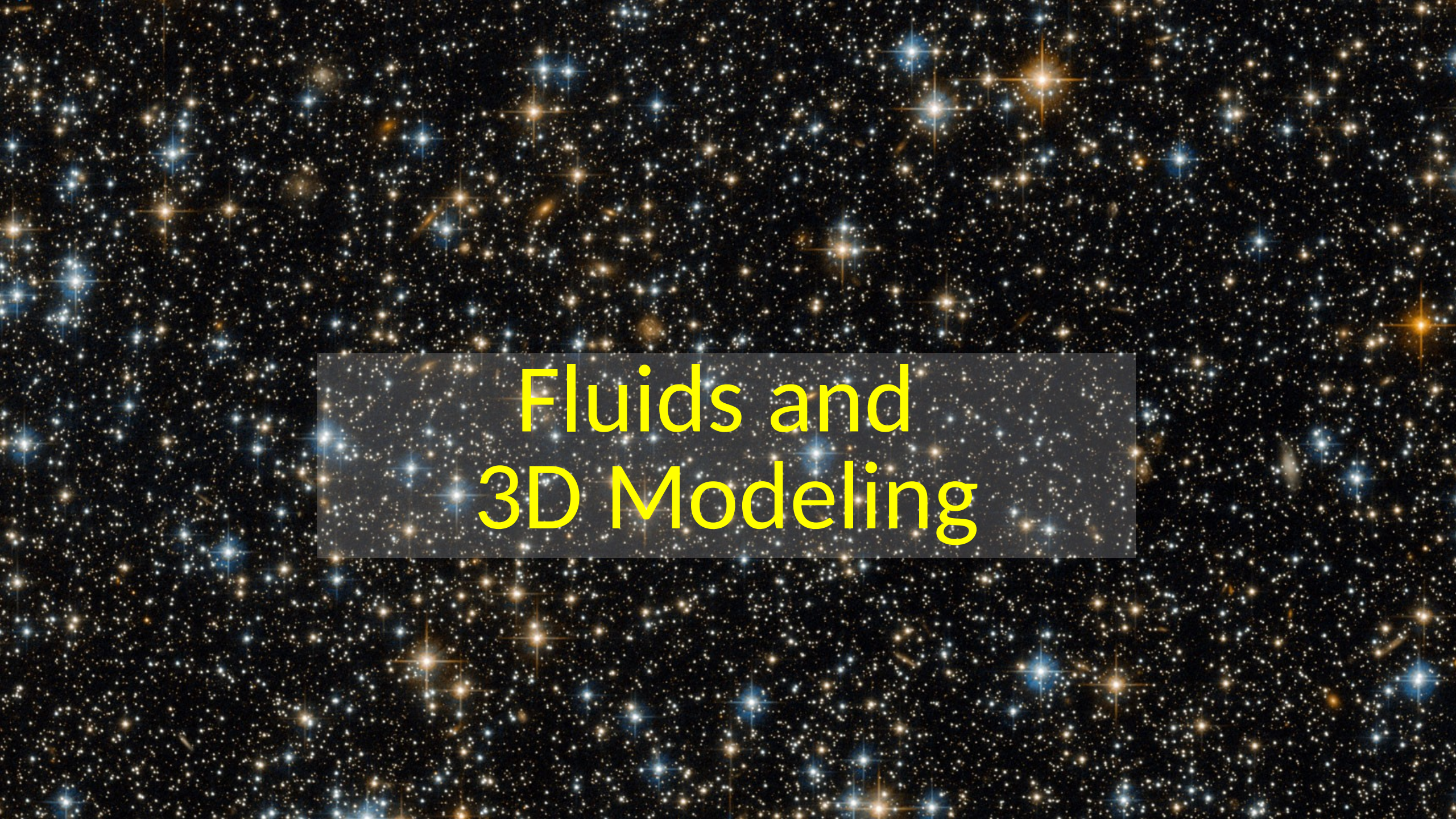




Fluids and 3D Modeling

Eulerian vs. Lagrangian Description

- We can imagine different reference frames when thinking about fluids
 - **Eulerian**: watch a particular region and look at how the fluid changes
 - Advection of new fluid and local sources will both contribute to the change at that location
 - **Lagrangian**: move along with the flow, tracking a particular fluid element
 - Properties in that fluid element change due to local sources, or compression of the fluid element

- Take a property $g(\mathbf{x}(t), t)$
- Eulerian description:
 - We are looking at a fixed $\mathbf{x}$, so we take

$$\frac{dg}{dt} = \frac{\partial g}{\partial t}$$

- Lagrangian description:
 - We move with the fluid:

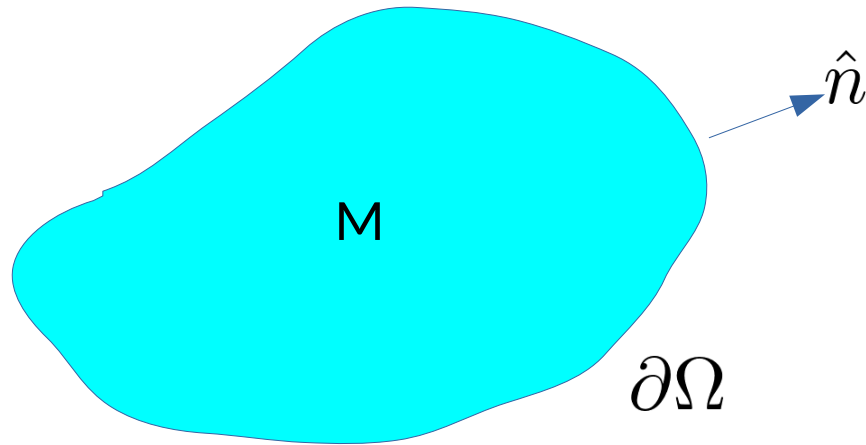
$$\frac{dg}{dt} = \frac{\partial g}{\partial t} + \frac{\partial g}{\partial x_j} \frac{dx_j}{dt} = \frac{\partial g}{\partial t} + \mathbf{v} \cdot \nabla g$$

- D/Dt will be the Lagrangian derivative:

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla$$

Mass Conservation

- Mass conservation



Ω is an arbitrary volume over which mass M is considered conserved.

The only changes to M will be due to the fluxes across the boundary

$$\begin{aligned}\frac{\partial}{\partial t} \int_{\Omega} \rho d\Omega &= - \int_{\partial\Omega} \mathbf{F}(\rho) \hat{\mathbf{n}} \cdot d\mathbf{S} \\ &= - \int_{\Omega} \nabla \cdot \mathbf{F}(\rho) d\Omega\end{aligned}$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

Mass Conservation

- In Lagrangian form:

$$\frac{D\rho}{Dt} = -\rho \nabla \cdot \mathbf{v}$$

- This tells us that density of a fluid element changes locally only due to compressive or expansive flow
- Note that if the *density of a fluid element is constant*, then

$$\nabla \cdot \mathbf{v} = 0$$

Momentum Conservation

- Start with Newton's 2nd law:

$$\underbrace{\rho dV}_{dM} \frac{D\mathbf{v}}{Dt} = \delta\mathbf{F}_{\text{body}} + \delta\mathbf{F}_{\text{surface}}$$

- Body forces (volumetric, e.g. gravity):

$$\delta\mathbf{F}_{\text{body}} = \rho\delta V \mathbf{f}$$

- Surface forces proportional to area (e.g. pressure):

$$\begin{aligned} (\delta\mathbf{F}_{\text{surface}})_i &= - \int_{\delta\Omega} P_{ij} dS_j \\ &= - \int_{\Omega} \frac{\partial P_{ij}}{\partial x_j} dV \end{aligned}$$

- Minus sign → p acts inward on element

- Together we have:

$$\rho \frac{Dv_i}{Dt} = \rho f_i - \frac{\partial P_{ij}}{\partial x_j}$$

- Ideal fluid → no viscosity:

$$P_{ij} = p\delta_{ij}$$

- Eulerian form:

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = \mathbf{f} - \frac{1}{\rho} \nabla p$$

Energy Conservation

- Start with 1st law of thermodynamics:

$$\frac{Dq}{Dt} = \frac{De}{Dt} + p \frac{D}{Dt} \left(\frac{1}{\rho} \right) = \frac{De}{Dt} + \frac{p}{\rho} \nabla \cdot \mathbf{v}$$

- This follows a fluid element
- Define heat rate / volume in the volume:

$$\mathcal{H} = \rho \frac{Dq}{Dt}$$

- e.g. thermal diffusion:

$$\mathcal{H} = -\nabla \cdot \mathcal{F} = \nabla \cdot (k \nabla T)$$

- Internal energy evolution:

$$\rho \frac{De}{Dt} + p \nabla \cdot \mathbf{v} = \mathcal{H}$$

- Total energy evolution → add $\mathbf{v} \cdot$ (momentum):

$$\frac{\partial(\rho E)}{\partial t} + \nabla \cdot [(\rho E + p)\mathbf{v}] = \mathcal{H} + \rho \mathbf{v} \cdot \mathbf{f}$$

- With:

$$E = e + \frac{1}{2} |\mathbf{v}|^2$$

- System closed with EOS

Euler Equations

- We have

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho v_i)}{\partial x_i} = 0$$

$$\frac{\partial(\rho v_j)}{\partial t} + \frac{\partial}{\partial x_i}(\rho v_i v_j + p) = \rho f_j$$

$$\frac{\partial(\rho E)}{\partial t} + \frac{\partial}{\partial x_i}[(\rho E + p)v_i] = \mathcal{H} + \rho v_i f_i$$

- In stars, we also carry a conservation equation for each species

$$\frac{\partial(\rho X_k)}{\partial t} + \frac{\partial}{\partial x_i}(\rho v_i X_k) = \rho \dot{\omega}_k$$

Conservative Form

- When we do numerical methods, we'll want to consider these equations in conservative form:

$$\mathbf{U}_t + \nabla \cdot \mathbf{F}(\mathbf{U}) = 0$$

- With

$$\mathbf{U} = \begin{pmatrix} \rho \\ \rho \mathbf{v} \\ \rho E \end{pmatrix}$$

- Introducing a momentum tensor: $T_{ij} = \rho v_i v_j + p \delta_{ij}$

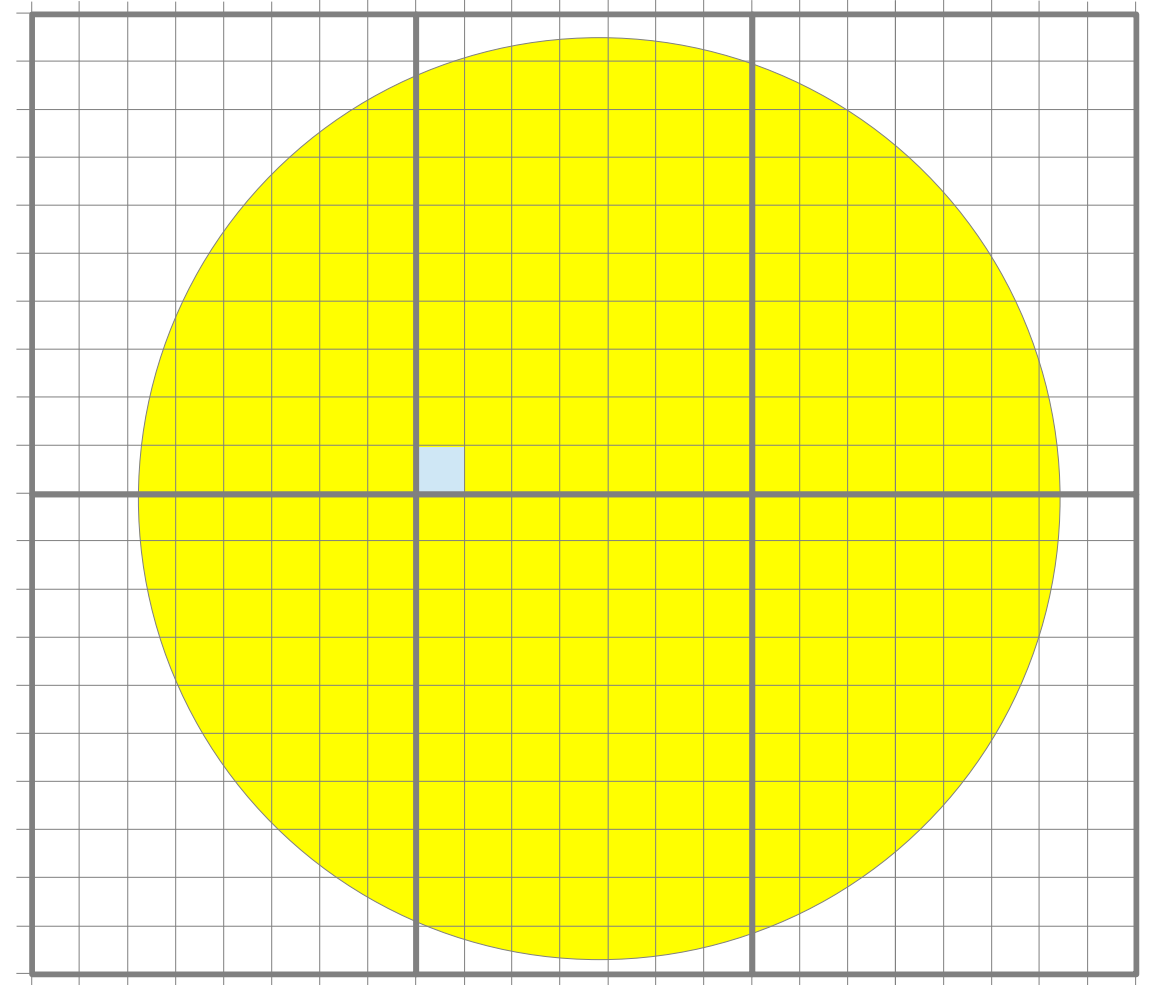
$$F_j = \begin{pmatrix} \rho v_j \\ T_{ij} \\ (\rho E + p)v_j - k \partial T / \partial x_j \end{pmatrix}$$

Computational modeling

- Solve conservation laws:

$$\mathcal{U}_t + \nabla \cdot \mathbf{F}(\mathcal{U}) = \mathbf{S}$$

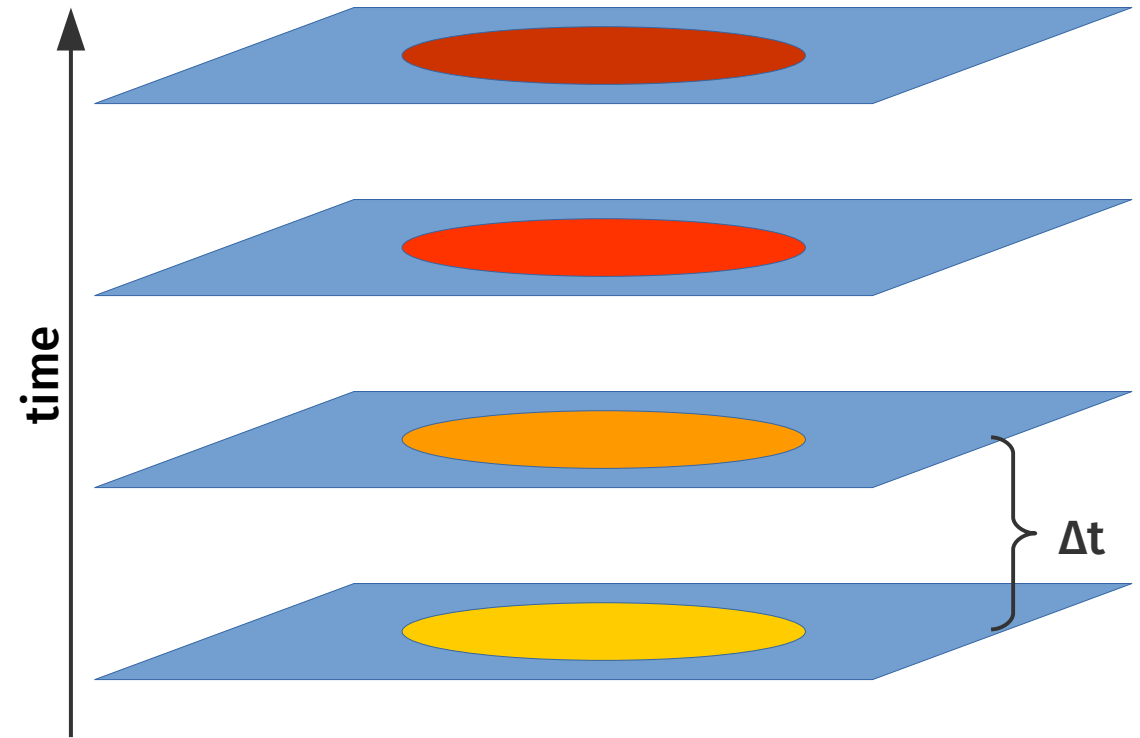
- Discretize by star on a grid
- Cell update depends its surroundings
 - Complexity is in computing the fluxes through boundaries
- Parallelism achieved by dividing domain across nodes



Computational modeling

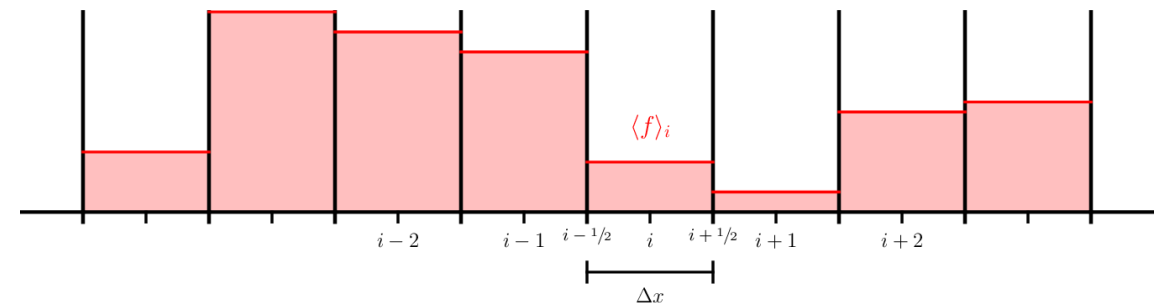
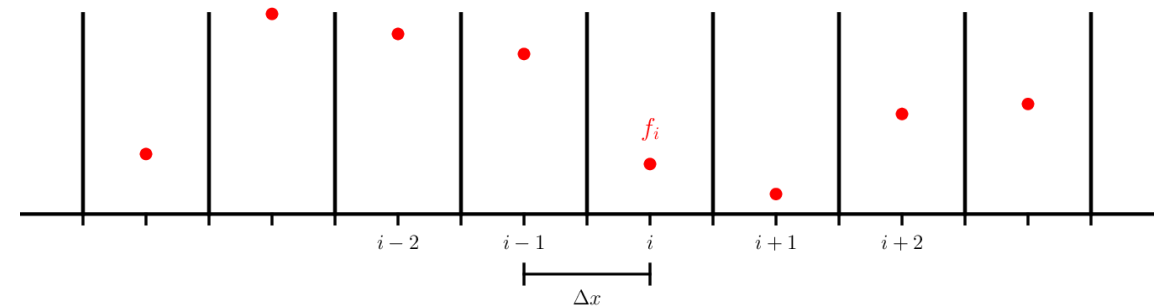
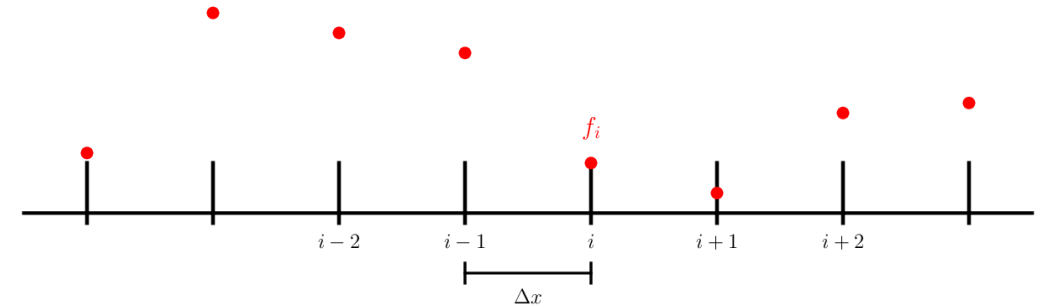
- Mathematical constraint on size of timestep
 - Information cannot move more than one zone per timestep
- Compressible hydro:

$$\Delta t \leq \frac{\Delta x}{|\mathbf{U}| + c_s}$$



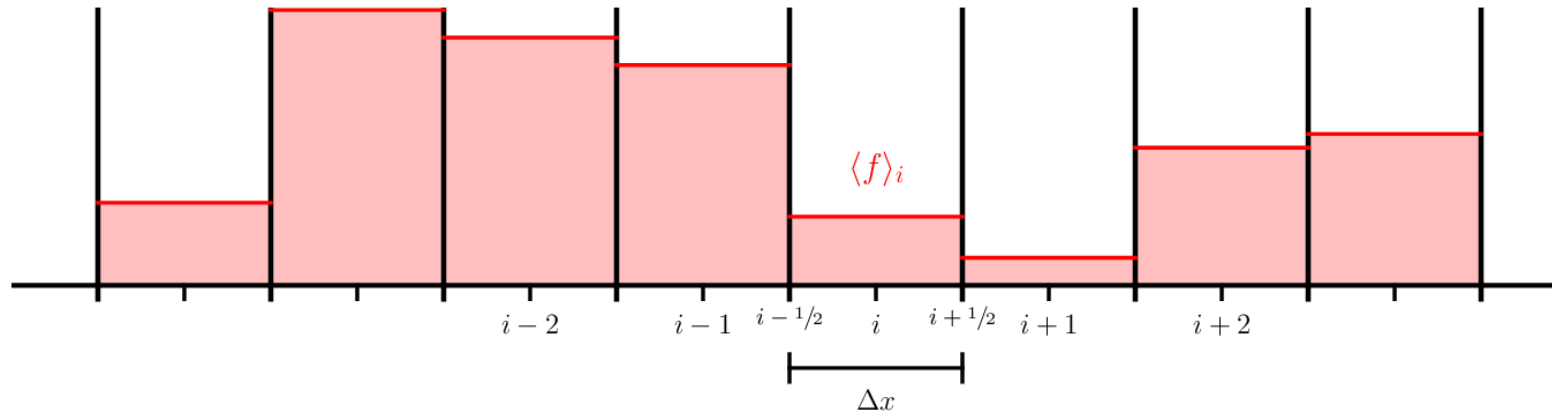
Discretization

- “Regular” finite-difference grid:
 - Data associated with nodes spaced Δx apart.
 - Note that here we can have a point exactly on the boundary
- Cell-centered finite-difference grid:
 - Cells of with Δx , data associated with a point at the center of the cell.
 - Note that data is $\Delta x/2$ inside the boundary
- Finite-volume grid:
 - Similar to the cell-centered grid $\rightarrow$ we divide the domain into cells/zones
 - Store the average value of the function in each zone.



Finite-Volume Approximation

- We store the average of a quantity in a zone



$$\langle f \rangle_i = \frac{1}{\Delta x} \int_{x_{i-1/2}}^{x_{i+1/2}} f(x) dx$$

Finite-Volume Approximation

- Integrate system over volume

$$\frac{1}{\Delta x} \int_{x_{i-1/2}}^{x_{i+1/2}} U_t dx = - \frac{1}{\Delta x} \int_{x_{i-1/2}}^{x_{i+1/2}} \frac{\partial}{\partial x} F(U) dx$$

- We get:

$$\frac{\partial}{\partial t} \langle U \rangle_i = - \frac{1}{\Delta x} \{ [F(U)]_{i+1/2} - [F(U)]_{i-1/2} \}$$

- Ensures conservation
- Now need to discretize in time

Time Discretization

- Method of lines

- After space discretization, we have an ODE in time:

$$\frac{\partial}{\partial t} \langle U \rangle_i = - \frac{\{[F(U)]_{i+1/2} - [F(U)]_{i-1/2}\}}{\Delta x}$$

- Construct flux using interface state:

$$[F(U)]_{i+1/2} = F(U_{i+1/2})$$

- Use an ODE method (like Runge-Kutta)

- Characteristic tracing:

- Predict state on interface at mid-point in time:

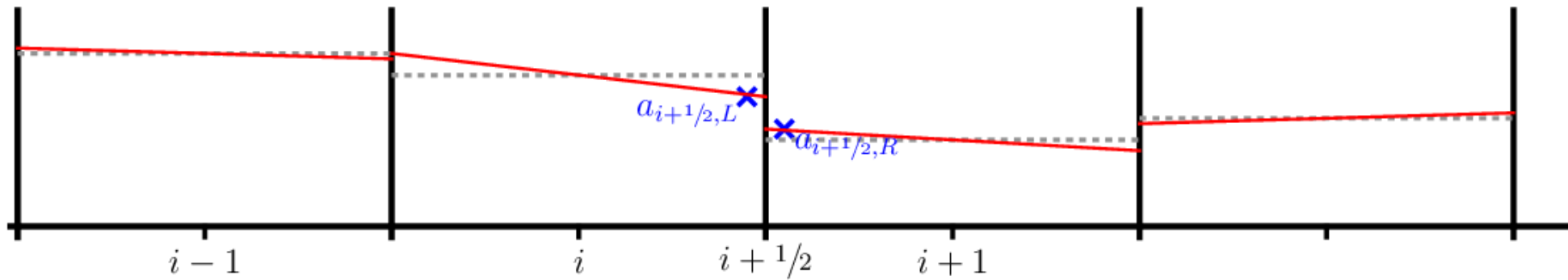
$$U_{i+1/2,L}^{n+1/2} = U_i + \frac{1}{2} \frac{\partial U}{\partial x} \Delta x + \frac{1}{2} \frac{\partial U}{\partial t} \Delta t + \dots$$

- Second-order update (time-centered):

$$\begin{aligned} \langle U \rangle_i^{n+1} &= \langle U \rangle_i^n \\ &+ \frac{\Delta t}{\Delta x} \left[F(U_{i-1/2}^{n+1/2}) - F(U_{i+1/2}^{n+1/2}) \right] \end{aligned}$$

Interface Prediction

- Going from cell-averages to interface values is done via reconstruction / prediction
- Reconstruction:
 - Fit a conservative polynomial to the cell average in a zone
 - Piecewise linear and piecewise parabolic polynomials are widely used
 - Limiting is needed to ensure that there are no overshoots
- Two different interface values are found (coming from left and right)

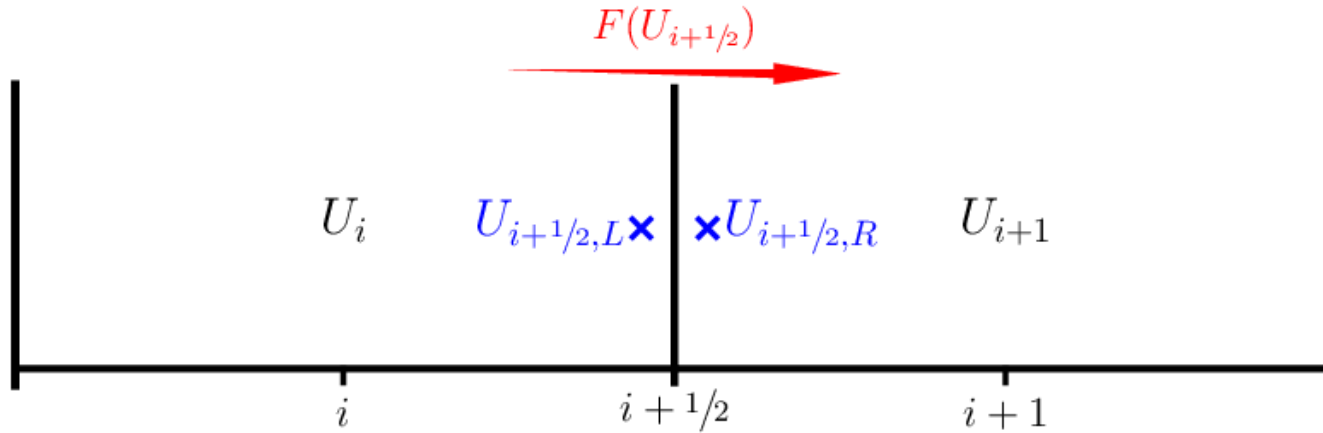


Riemann Problem

- We have two predictions for the interface
- Solve:

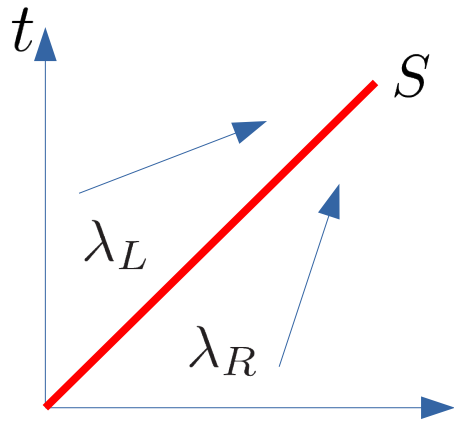
$$U_{i+1/2} = \mathcal{R}(U_{i+1/2,L}, U_{i+1/2,R})$$

- Requires detailed understanding of the characteristic structure of the system
 - Euler equations have 3 characteristic waves
 - Only density jumps across the middle wave (contact discontinuity)
- Note: usually work in terms of primitive es, $\mathbf{q}$

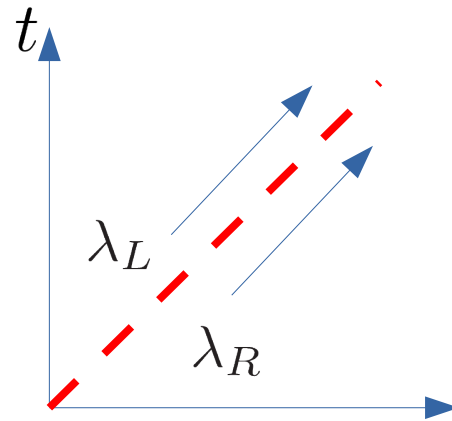


Riemann Problem

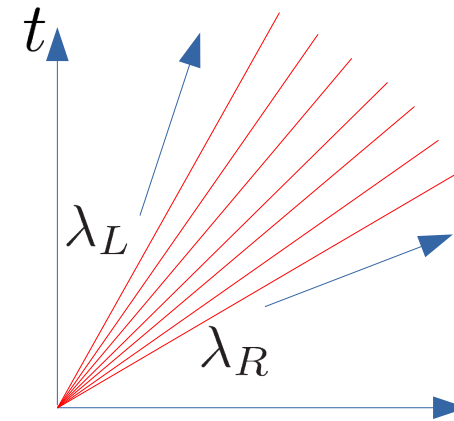
- Wave type is determined by whether the characteristics converge or diverge



Shock wave



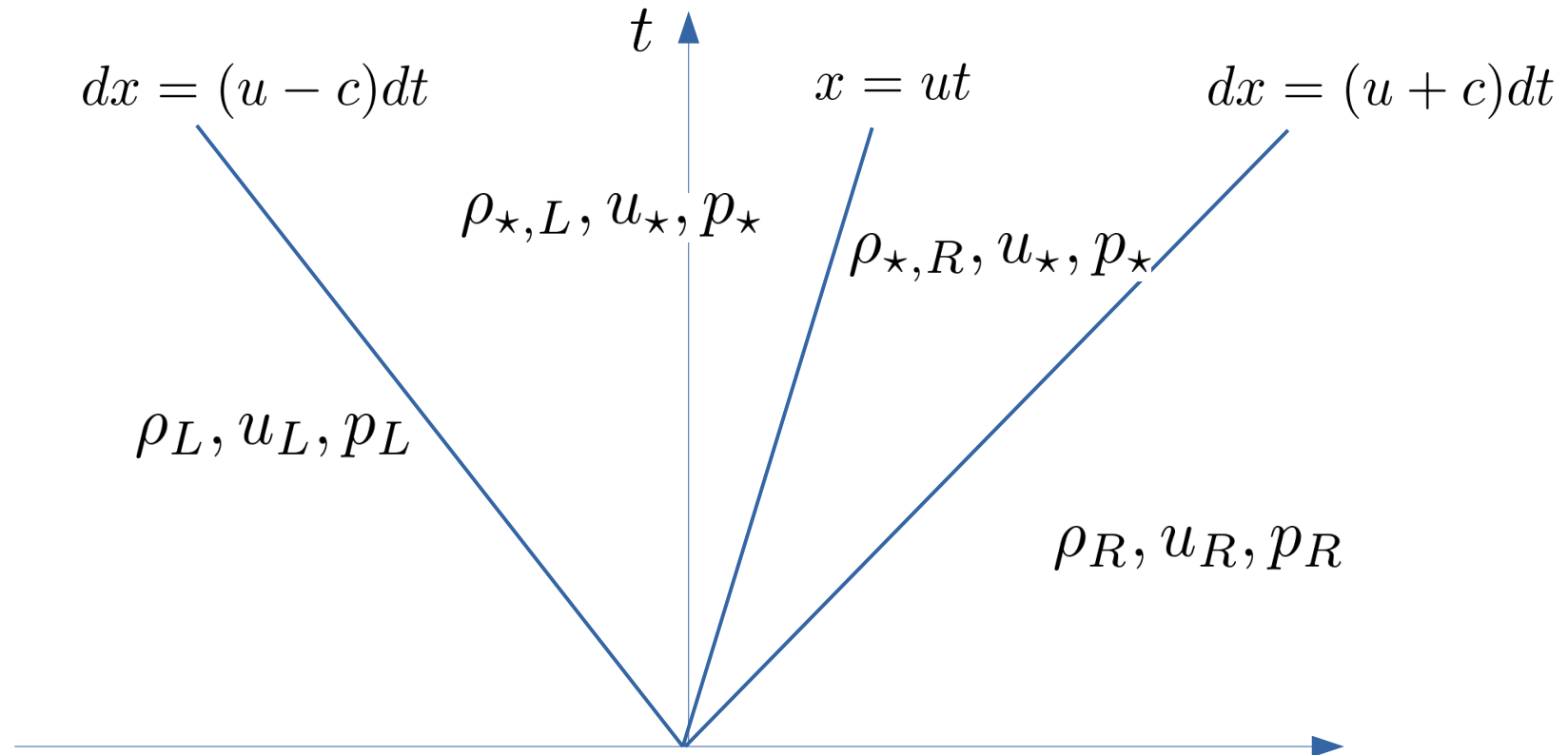
Contact Discontinuity



Rarefaction

Riemann Problem

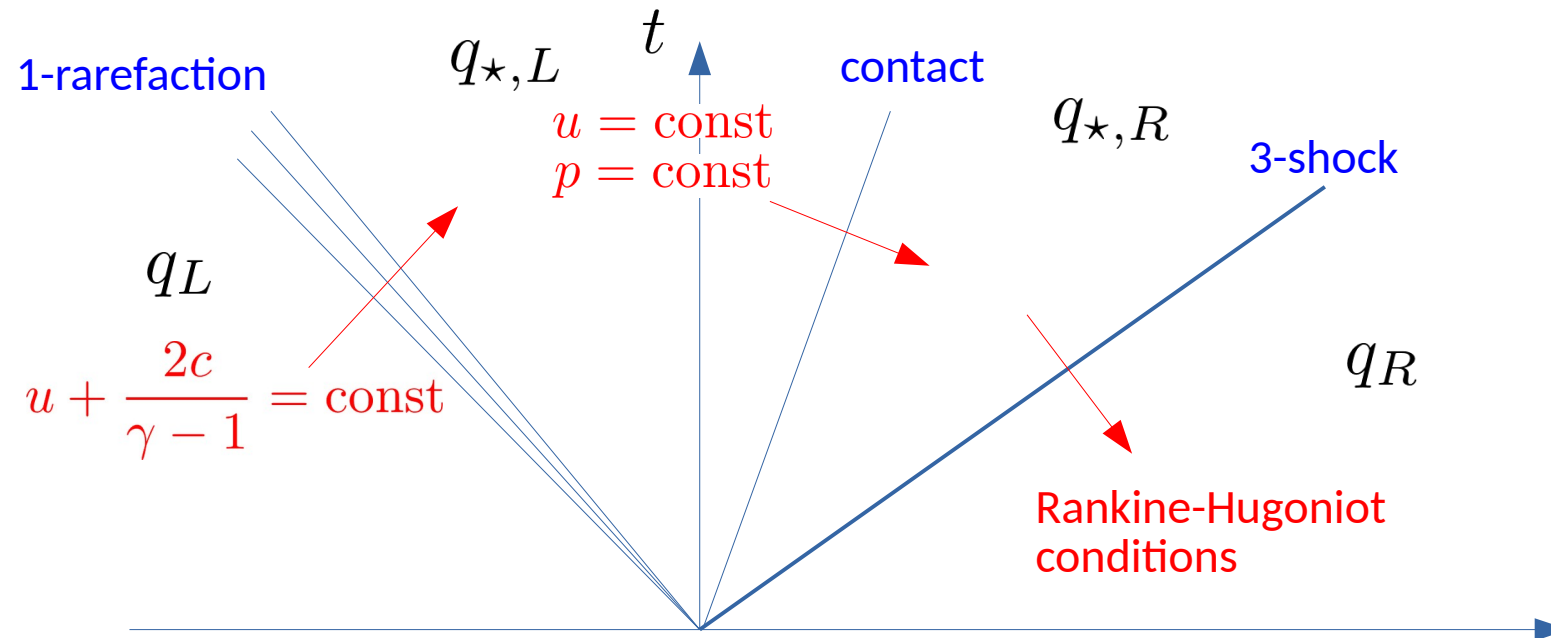
- Riemann problem for Euler equations appears as:



- Solution requires finding 4 quantities: $\rho_{\star,L}, \rho_{\star,R}, u_{\star}, p_{\star}$

Riemann Problem

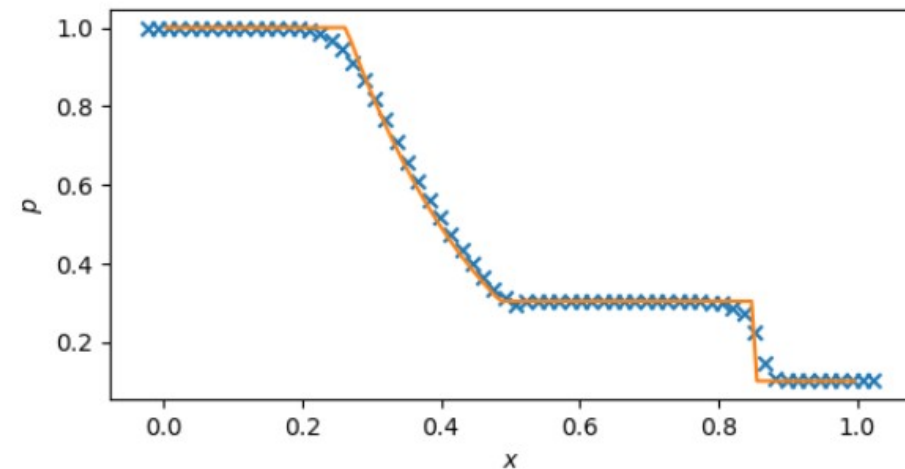
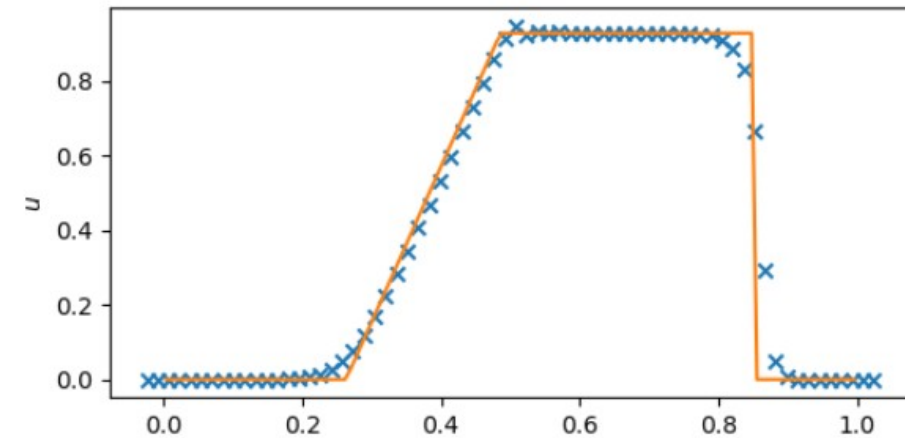
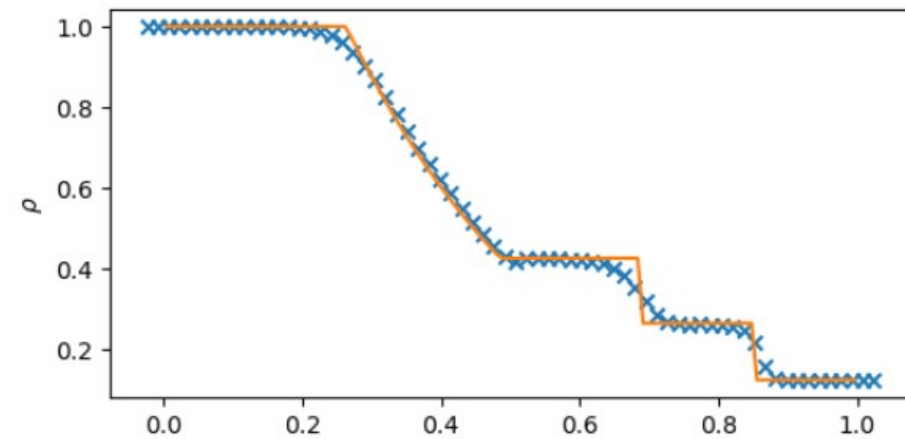
- Example wave structure:



See https://zingale.github.io/computational_astrophysics/euler/euler-riemann.html

Example Solution

- Sod problem is a common test
 - Exhibits all 3 types of waves



Flowchart

- Initialize grid
 - Set initial conditions
 - Time evolution loop:
 - Compute timestep
 - Reconstruct state to interfaces
 - Solve Riemann problem at each interface
 - Do conservative update
 - Output
- For a simple 1D example see my notes here:
https://zingale.github.io/computational_astrophysics/euler/euler.html
 - For a variety of 2D algorithms, check out my pyro code:
<https://python-hydro.github.io/pyro2/>

Reformulations

- Compressible codes limited by soundspeed:

$$\Delta t = \min \left\{ \frac{\Delta x}{|u| + c} \right\}$$

- For $M \ll 1$:

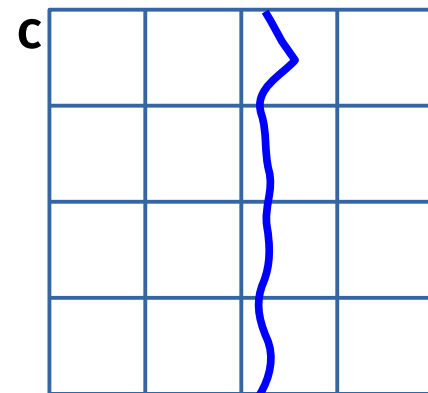
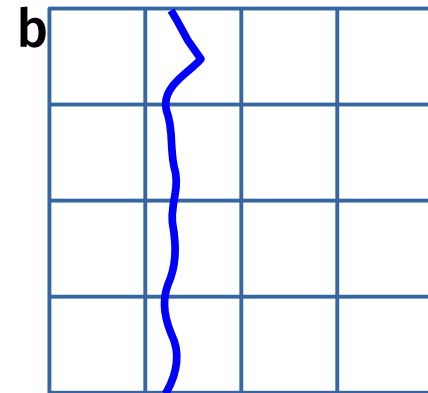
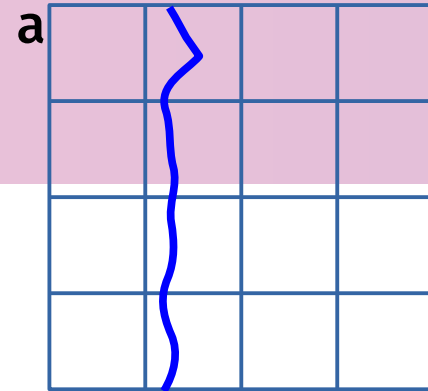
$$\Delta t \approx \Delta x / c$$

- Very low-Mach number flows takes $\sim 1/M$ steps to move a single zone

- We want:

$$\Delta t \approx \Delta x / |u|$$

- Low-speed methods exist to filter soundwaves



► A Mach 0.01 front moving to the right (a) initially, (b) after 1 step, (c) after 100 steps.

Popular Low Speed Approximations

- Incompressible

- No compressibility effects modeled

$$\nabla \cdot U = 0$$

- Anelastic

- Compressibility due to stratified atmosphere
- Small thermodynamic perturbations from a static hydrostatic (usually isentropic) background

$$\nabla \cdot (\rho_0 U) = 0$$

- Low Mach number combustion

- Local compressibility due to heat release and diffusion
- Large temperature/density jumps permitted

$$\nabla \cdot U = S$$

- Pseudo-incompressible

- Both compressibility due to background stratification and local heat release
- Static background
- Originally: ideal gas EOS

$$\nabla \cdot (p_0^{1/\gamma} U) = S$$

Low Speed Stellar Flows

- Developed a new algorithm for low speed stellar flows
 - Recasts the equation of state as an elliptic constraint on velocity
 - Decouples soundwaves from system
- Single step is more expensive than compressible
 - Many fewer steps needed
 - Overall wallclock time is dramatically lower

